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Beyond Completion: Probing Cumulative State Tracking to Predict LLM Agent Performance

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Abstract

Task-completion rate is the standard proxy for LLM agent capability, but models with identical completion scores can differ substantially in their ability to track intermediate state. We introduce Working Memory Fidelity-Active Manipulation (WMF-AM), a calibrated no-scratchpad probe of cumulative arithmetic state tracking, and evaluate it on 20 open-weight models (0.5B–35B, 13 families) against a released deterministic 10-task agent battery. In a pre-specified, Bonferroni-corrected analysis, WMF-AM predicts agent performance with Kendall's τ=0.612 (p<0.001, 95 % CI [0.360, 0.814]); exploratory partial-τ analyses suggest this signal persists after controlling for completion score and model scale. Three construct-isolation ablations (K=1 control, non-arithmetic ceiling, yoked cancellation) support the interpretation that cumulative state tracking under load, rather than single-step arithmetic or entity tracking alone, is the primary difficulty source. K-calibration keeps the probe in a discriminative range where prior fixed-depth benchmarks become non-discriminative; generalization beyond this open-weight sample remains open.

1 Introduction

This paper investigates whether a controlled probe of cumulative arithmetic state tracking captures variance in downstream agent performance not reflected by completion alone. Task-completion metrics are useful and widely applicable, but completion scores may compress variance in process-level capabilities that matter for reliable multi-step agent behavior. We call this the Completion Fallacy: the inference that completion success alone is sufficient for inferring latent process competence. Benchmarks such as HotpotQA [Yang et al., 2018], MMLU [Hendrycks et al., 2021a], and AgentBench [Liu et al., 2023] measure what LLMs accomplish; process-sensitive probes ask how models process information under load, a complementary evaluation lens.

We introduce and empirically validate WMF-AM, a controlled probe for cumulative arithmetic state tracking under load, and situate it within a working interpretive taxonomy, the Cognitive Evaluation Framework (CEF). WMF-AM is distinguished from prior state-tracking work [Kim and Schuster, 2023, Huang et al., 2025, Xia et al., 2025] by cumulative arithmetic load manipulation, targeted ablation controls, and downstream predictive validity on a deterministic agent battery; Section 2 details the comparison.

This paper makes four contributions. (1) Construct and probe. We formalize cumulative arithmetic state tracking under load as a measurable process-level capability and introduce

WMF-AM, a K-calibrated, no-scratchpad probe for evaluating it (Section 3). (2) Predictive validity. Across 20 open-weight models (0.5B–35B, 13 families), WMF-AM predicts performance on a non-overlapping deterministic 10-task agent battery (τ=0.612, p<0.001, 95 % CI [0.360, 0.814]; pre-specified, Bonferroni-corrected; Section 4). Exploratory partial-τ analyses suggest the signal persists after controlling for task completion and model scale. (3) Construct isolation. Three targeted ablations, a K=1 single-step control, a non-arithmetic ceiling condition, and a yoked cancellation control, support the interpretation that cumulative state tracking under load, rather than single-step arithmetic or generic entity tracking, is the primary difficulty source within this sample (Section 4.2). (4) Measurement robustness. WMF-AM rankings are stable across random seeds, prompt templates ($\tau_{\text{bare,chat}}$=0.524, p=0.006; pre-specified), natural-language paraphrases, and leave-one-family-out analyses (Section 4.4). A glossary of all acronyms appears in Table 11 (Appendix F). Generalization to closed API models and broader criteria remains open; the CEF taxonomy is an interpretive framework, not an empirical claim.

2 Related Work

Completion-based evaluation and the process-quality gap. Standard benchmarks, MMLU [Hendrycks et al., 2021a], HotpotQA [Yang et al., 2018], AgentBench [Liu et al., 2023], HELM [Liang et al., 2023], measure outcome correctness without probing the reasoning process. This creates what we term the Completion Fallacy, an instance of multiple realizability [Putnam, 1967]: identical outcomes produced by categorically different processes, documented also as underspecification [D'Amour et al., 2022] and shortcut learning [Geirhos et al., 2020]. Within our 20-model sample, 30/105 pairs with $|\Delta\text{completion}| \leq 0.05$ show $|\Delta\text{WMF-AM}| \geq 0.10$, illustrating that, in this sample, completion compresses process-level variance.

Evidence of process-outcome dissociation. Recent work documents process-outcome gaps at scale. PAE [Xie et al., 2026] finds 27–78% of agent successes involve corrupted trajectories; Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] show correct answers produced despite broken reasoning chains; Turpin et al. [Turpin et al., 2023] demonstrate unfaithful chain-of-thought explanations. AgentProcessBench [Fan et al., 2026] provides 8,509 human-annotated steps showing step-level quality diverges from outcomes. Process supervision [Lightman et al., 2023, Uesato et al., 2022] trains on step-level labels; CEF measures process quality without requiring labelled training data.

State-tracking and process-sensitive probes. A growing body of work probes how LLMs maintain and update internal state representations. Entity Tracking [Kim and Schuster, 2023] tests passive state retrieval (e.g., which container holds a given object after a sequence of swaps), while Latent State Persistence [Huang et al., 2025] extends to sequential manipulation of hidden integers—the closest prior paradigm to WMF-AM, though without arithmetic accumulation or downstream agent-predictive validation. Several recent benchmarks [Rezaee et al., 2025, Li et al., 2025] similarly probe object or entity state, and Zhang et al. [Zhang et al., 2024] motivate working-memory-style evaluation, but none pair ablation controls with an external criterion battery. Minerva [Xia et al., 2025] includes a Quantity State task that is structurally closest to WMF-AM, sequential additive/subtractive operations, yet its fixed depth of K=200 places all 20 of our study models at exactly 0.000, rendering the probe non-discriminative (τ undefined).

WMF-AM addresses this gap along three axes: K-calibration restoring discriminability across the 0.5B–35B range, a construct-isolation ablation suite, and predictive validity against a deterministic agent battery (head-to-head with Minerva in Section 5).

3 WMF-AM and Its Interpretive Framework (CEF)

CEF (Cognitive Evaluation Framework) is a design lens that organizes process-sensitive probes into four proposed dimensions: WMF (Working Memory Fidelity), MCC (Metacognitive Calibration), EMC (Episodic Memory Coherence), and CLA (Cognitive Load Adap-

tation), with 11 sub-dimensions. This dimensional structure is hypothesized; factor analysis requires $N \geq 30$ (Appendix C; glossary in Table 11).

Flagship probe: WMF-AM. WMF-AM (Active Manipulation) is a probe of cumulative arithmetic state tracking under load, inspired by working-memory span paradigms. An initial numeric state is presented, K sequential additive/subtractive operations arrive, and the model must report the final cumulative state. The pilot uses depths $K \in \{3, 5, 7\}$ (dissociation study) and $K \in \{2, 4, 6, 8, 12\}$ (yoked cancellation control), across three surface forms (points scoring, warehouse inventory, bank accounts). Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential arithmetic transformation under load.

Scope. WMF-AM measures difficulty with maintaining a running cumulative sum across K operations. A K=1 control separates cumulative tracking from single-step arithmetic: 13/15 models achieve near-ceiling accuracy at K=1 ($\bar{x}$=0.957) versus $\bar{x}$=0.151 at K=7, indicating that cumulative load, not arithmetic skill per se, is the primary difficulty driver.

Worked example (K=3, points form): "Alice starts with 10 points. Alice gains 5 points. Alice loses 3 points. Alice gains 7 points. What is Alice's current score?" Correct answer: 19. Scoring: exact match (1 if the model outputs 19, 0 otherwise). A model that tracks each update cumulatively ($10 \rightarrow 15 \rightarrow 12 \rightarrow 19$) succeeds; a model that performs only the last operation or guesses a plausible number fails. At K=7, the chain length taxes state maintenance, producing the diagnostic depth×accuracy interaction.

Yoked control example (K=3): "Alice starts with 10 points. Alice gains 5 points. Alice loses 5 points. Alice gains 3 points. Alice loses 3 points. Alice gains 7 points. Alice loses 7 points. What is Alice's current score?" Correct answer: 10 (initial state). Each operation is immediately followed by its inverse, requiring identical arithmetic but no cumulative tracking.

Preliminary pilots for MCC, EMC, and CLA are reported in Appendix N–O; construct-validity framework and falsification criteria are in Appendix I.

4 Validation Study: WMF-AM Provides Initial Predictive Evidence for Agent Performance

Variable definitions. Three scores are used throughout this section and should not be confused. WMF-AM Score: the probe output, mean accuracy over $K \in \{3, 5, 7\}$ depths, 4 random seeds, and 3 surface forms (points / inventory / accounts) for each model, all collected under each model's Ollama default chat template (Section 3). Prompt-wrapper variation (bare / chat / CoT) is analyzed separately as a robustness check in Phase 5 (Section 4.2) and is not folded into the primary WMF-AM Score. (In Phase 5, "chat" denotes the same Ollama default chat wrapper used for the primary evaluation; "bare" strips it.) Outcome Correctness (OC): the 100-item general-knowledge completion battery used as a general-capability baseline (Section 4, Phase 1); this is the "completion score" referred to throughout. Agent Battery Score (ABS): the 10-task deterministic downstream criterion (Phase 6); this is the "agent score" or "downstream target" used in all predictive validity analyses. OC and WMF-AM serve as predictors; ABS is the downstream criterion.

Statistical methodology. All rank correlations use Kendall's τ-b (ties accounted for). Partial τ uses Somers residualization [Somers, 1962]. Bootstrap 95% CIs: 10,000 model-level and family-clustered resamples; family-clustered CI $[0.341, 0.815]$ is virtually identical to model-level $[0.360, 0.814]$, no single family is load-bearing ($\tau \in [0.503, 0.649]$ across leave-family-out runs). Pre-specified primary analyses (Bonferroni-corrected, α=0.025): (1) predictive validity ($\tau = 0.612$) and (2) template stability ($\tau = 0.524$). Both were frozen before expansion-set collection; all other analyses are exploratory.

Formal criterion. Let $Y_i \in [0, 1]$ denote the outcome score (OC) and $Z_i \in [0, 1]$ a process-sensitive measure (WMF-AM) for model i. Outcome metrics are statistically insufficient w.r.t. a downstream criterion D if partial $\tau(Z, D \mid Y) > 0$ at conventional α. This condition

is benchmark-specific, not a universal claim. We report an exploratory test of this criterion in Section 4.1.

Table 1. Study map: complete analysis register. All τ = Kendall's τ-b. C = pre-specified primary (Bonferroni-corrected at α=0.025); E = exploratory (hypothesis-generating only; require held-out replication before drawing strong conclusions). Predictor: WMF-AM Score; Criterion: Agent Battery Score (ABS); Baseline: Outcome Correctness (OC).

Relation	Subset	τ	N	p	95% CI	Type
WMF-AM → Agent overall	All models	0.612	20	<0.001	$[0.360, 0.814]$	C
Bare ↔ Chat template ranks	15 models	0.524	15	0.006	—	C
WMF-AM → Agent (partial | OC)	All models	0.411	20	0.011	—	E
WMF-AM → Agent (partial | log params)	All models	0.503	20	<0.01	—	E
WMF-AM → Agent (partial | OC + log params)	All models	0.389	20	0.016	—	E
WMF-AM → Agent non-WMF (7 tasks)	All models	0.623	20	<0.001	—	E
WMF-AM → Agent WMF-heavy (3 tasks)	All models	0.527	20	0.003	—	E
WMF-AM ↔ OC	15 models	0.279	15	0.150	$[-0.156, 0.637]$	E
CONV-RFPOC ↔ AGENT-PQ	7 models	0.905	7	0.003	—	E

Note on sample sizes. Different analyses use different subsets: N=20 for all predictive validity and completion-battery analyses (the full study set); N=15 for WMF-AM template stability and divergent validity (original 7 + expansion 8; before small models were added); N=7 for convergent validity only (original models with AGENT-PQ and CONV-RFPOC data collected). AGENT-PQ is a separate LLM-judged process-quality measure used only for convergent validity and should not be confused with ABS (the deterministic exact-match agent battery used as the primary downstream criterion). All sample sizes are reported explicitly at each result.

Twenty open-weight models from thirteen architectural families form the study set: seven original models (Qwen 7B/14B/32B, Llama 3.1:8b, Gemma 2:27b, DeepSeek-R1:14b, Mistral:7b), eight expansion models (Phi-3:14b, Gemma-2:9b, Qwen-2.5:3b, Llama-3.2:3b, DeepSeek-R1:7b, Mixtral:8x7b, Command-R:35b, Yi:34b), and five small baseline models (0.5B–2B parameters; Table 3).

Models were evaluated in six phases. Phase 1 (Outcome Correctness) used a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, temperature = 0. Phase 2 (WMF-AM) administered 15 probes at depths $K \in \{3, 5, 7\}$ with 4 independent random seeds for all 20 models. Phase 3 (MCC-MA) applied a 20-problem self-monitoring probe. Phase 4 (Yoked Cancellation Control) administered 20 trials per depth at $K \in \{2, 4, 6, 8, 12\}$ using operations that cancel (gain then lose, transfer then reverse), so the correct answer equals the initial state despite requiring identical arithmetic parsing as WMF-AM. Phase 5 (Template Harmonization) re-administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) to test prompt-format sensitivity. Phase 6 (Agent Validation) administered 10 multi-step agent tasks (tool use, multi-step reasoning, state tracking) to all 20 models to test predictive validity. All models are Ollama-hosted open-weight variants using default chat templates, fp16 quantization where available, and greedy decoding (temperature = 0, no top-p sampling). No task overlap exists between phases (full study matrix in Appendix K). Table 2 summarizes the experimental design.

4.1 Primary Confirmatory Results

WMF-AM predicts Agent Battery Score. Figure 1 and Table 3 present the key finding across N=20 models from 13 families. WMF-AM (4-seed means) spans 0.050–0.983 ($\Delta = 0.933$) while the 100-item completion battery spans 0.44–0.92 ($\Delta = 0.480$). Among the 15 mid-to-large models (≥3B), OC clusters at 0.75–0.92 (SD = 0.052) while WMF-AM spans 0.133–0.983 (SD = 0.228): OC masks substantial process-quality differences. Predictive validity: WMF-AM significantly predicts performance on a deterministic 10-task multi-step agent battery within this 20-model open-weight sample ($\tau = 0.612$, $p < 0.001$, N=20; bootstrap 95% CI $[0.360, 0.814]$). A held-out check on llama3.1:70b (withheld from the study set) yields $\tau = 0.627$ ($p = 0.0002$, N=21), essentially unchanged; this is consistent with but does not constitute replication. Exploratory incremental validity: partial $\tau = 0.411$ ($p = 0.011$)

Table 2. Study design summary. All phases: temperature = 0, greedy decoding, Ollama-hosted models with default chat templates. All 20 models evaluated with 4 seeds for WMF-AM.

Phase	Probe	Depths	Items	Seeds	Scoring
1. Outcome	100-item battery (5 domains)	—	100	1	Exact match (0/1)
2. WMF-AM	State tracking	K=3, 5, 7	15/depth	4	Exact match on final state
3. MCC-MA	Self-monitoring	—	20	1	Pearson r (predicted vs actual)
4. Yoked	Cancellation ops	K=2, 4, 6, 8, 12	20/depth	1	Exact match on initial state
5. Template	3 prompt wrappers	K=3, 5, 7	15/depth	1	Rank stability (τ)
6. Agent	10 multi-step tasks	—	10	1	Deterministic task score (0–1)

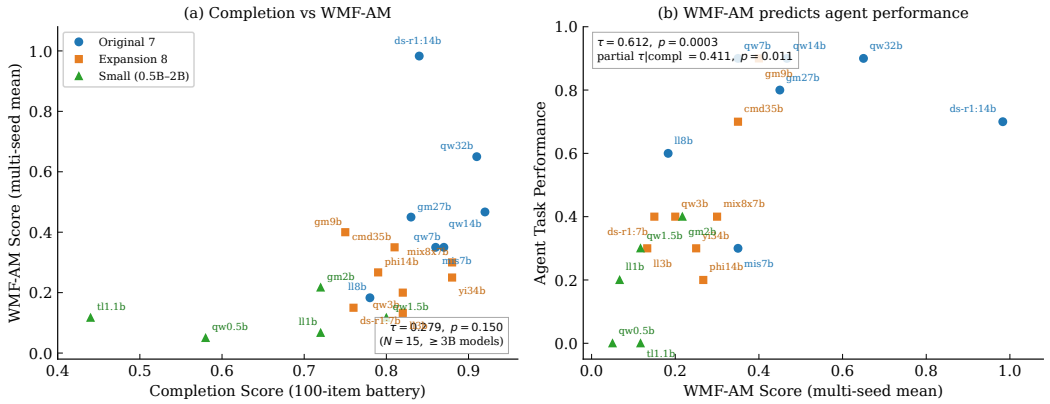

Figure 1. (a) Completion clusters while WMF-AM spans a wide range (N=20, 13 families). (b) WMF-AM predicts downstream agent performance (τ=0.612, p<0.001; pre-specified confirmatory). Exploratory: partial τ=0.411 (p=0.011) after controlling for completion (same sample; requires held-out replication). Blue circles = original 7; orange squares = expansion 8; green triangles = small models (0.5B–2B).

after controlling for OC, and partial $\rho = 0.437$ ($p = 0.008$) via Spearman (robustness check), are consistent across two rank-correlation families (exploratory; same N=20 sample). OC also predicts agent performance ($\tau = 0.428$, $p = 0.012$). The primary pre-specified evidence is twofold: (1) significant pre-specified predictive validity ($\tau = 0.612$) and (2) significant pre-specified template stability ($\tau = 0.524$, $p = 0.006$).

4.2 Construct Isolation and Alternative Explanations

Completion battery design. Phase 1 uses a custom 100-item battery (five domains × five difficulty levels; score resolution 0.01) as a within-study general-capability control, not a standard benchmark suite. An exploratory log(parameter count) control ($\tau = 0.503$, $p < 0.01$) provides a complementary proxy independent of battery choice; the joint partial τ (0.389, $p = 0.016$) controls for both simultaneously. With the addition of 5 small models (0.5B–2B), OC ranges from 0.44 to 0.92 (SD = 0.131, N=20) and correlates with agent performance ($\tau = 0.428$, $p = 0.012$); WMF-AM shows an exploratory signal of incremental prediction (partial $\tau = 0.411$, $p = 0.011$). Among the 15 mid-to-large models (≥3B), OC remains compressed (SD = 0.052), consistent with range restriction being a property of the model population rather than the battery design.

Exploratory architectural observations (Appendix U). Only deepseek-r1:14b shows positive WMF−Yoked residual (+0.173), suggesting genuine cumulative tracking; all other 14 models show negative residuals, with the largest gaps in smaller models. Figure 2 shows depth-dependent degradation curves (N=15).

Alternative explanations and template sensitivity. WMF-AM scores may reflect nuisance factors beyond state tracking: instruction-following ability, tokenizer effects on numeric parsing, scratchpad/chain-of-thought policy differences, or prompt-format sensitivity. Phase 5 (template harmonization) directly tests the last concern: across 15 models × 3 templates, model rankings under bare and chat wrappers are significantly correlated ($\tau_{\text{bare,chat}} = 0.524$,

Model	Family	Outcome	WMF-AM	Agent	Yoked	WMF−Yoked
Original 7 (all 4-seed WMF-AM)						
deepseek-r1:14b	DeepSeek	0.840	0.983	0.70	0.810	+0.173
qwen2.5:32b	Qwen	0.910	0.650	0.90	1.000	−0.350
qwen2.5:14b	Qwen	0.920	0.467	0.90	0.710	−0.243
gemma2:27b	Gemma	0.830	0.450	0.80	0.620	−0.170
qwen2.5:7b	Qwen	0.870	0.350	0.90	0.770	−0.420
mistral:7b	Mistral	0.860	0.350	0.30	0.550	−0.200
llama3.1:8b	Llama	0.780	0.183	0.60	0.340	−0.157
Expansion 8 (all 4-seed WMF-AM)						
gemma2:9b	Gemma	0.750	0.400	0.90	0.960	−0.560
command-r:35b	Cohere	0.810	0.350	0.70	0.920	−0.570
mixtral:8x7b	Mistral	0.880	0.300	0.40	1.000	−0.700
phi3:14b	Phi	0.790	0.267	0.20	0.950	−0.683
yi:34b	Yi	0.880	0.250	0.30	1.000	−0.750
qwen2.5:3b	Qwen	0.820	0.200	0.40	0.970	−0.770
deepseek-r1:7b	DeepSeek	0.760	0.150	0.40	0.940	−0.790
llama3.2:3b	Llama	0.820	0.133	0.30	0.510	−0.377
Small models (broader baseline)						
gemma2:2b	Gemma	0.720	0.217	0.40	—	—
qwen2.5:1.5b	Qwen	0.800	0.117	0.30	—	—
tinyllama:1.1b	TinyLlama	0.440	0.117	0.00	—	—
llama3.2:1b	Llama	0.720	0.067	0.20	—	—
qwen2.5:0.5b	Qwen	0.580	0.050	0.00	—	—
Δ	—	0.480	0.933	0.90	—	—

Table 3. Per-model scores: WMF-AM Score, Outcome Correctness (OC), Agent Battery Score (ABS), and Yoked control accuracy (N=20, 13 families). WMF-AM Score = 4-seed mean at $K \in \{3, 5, 7\}$; OC = 100-item general-knowledge battery (Outcome Correctness; baseline predictor); ABS = 10-task deterministic agent battery (downstream criterion; pre-specified primary analysis). WMF-AM (pre-specified primary): τ=0.612 (p<0.001, confirmatory, C). OC → ABS: τ=0.428 (p=0.012, exploratory, E). Yoked column: cancellation-control accuracy (original 15 models only).

$p = 0.006$), indicating that WMF-AM rankings are moderately robust to prompt-format variation. Chain-of-thought (CoT) templates push nearly all models to ceiling (≥ 0.93), eliminating variance.

Construct boundary. WMF-AM probes internalized state tracking under the no-scratchpad regime; the CoT ceiling (≥ 0.93) is a measurement boundary, not a validity threat, relevant to deployment contexts where intermediate outputs are unmonitored [Nye et al., 2021]. Predictive validity is established within this regime; generalization to CoT-enabled or tool-augmented settings is untested. The moderate bare↔chat stability ($\tau = 0.524$) indicates residual prompt sensitivity; instruction-following and tokenizer effects remain live confounds. Ablation results (N=15). (i) Non-arithmetic ceiling. Replacing numeric accumulation with direct-assignment updates (color, location, status; 3 domains $\times$ $K \in \{3, 5, 7\}$) raises accuracy to near-ceiling (mean 0.98, range 0.91–1.00) versus the arithmetic range of 0.20–0.72, supporting the interpretation that WMF-AM difficulty is driven by cumulative arithmetic load rather than generic entity tracking. The K=1 control further separates cumulative from single-step arithmetic: 13/15 models achieve $\bar{x}$=0.957 at K=1 yet $\bar{x}$=0.151 at K=7. Multi-step serial-composition alternatives (e.g., error propagation) cannot be excluded without mechanistic analysis. (ii) Prompt paraphrase (5 templates: original, formal, casual, minimal, verbose): model rankings are stable across natural-language rephrasings; mean cross-template Kendall $\tau = 0.54$ (5/6 pairs $p < 0.05$); original-vs-formal $\tau = 0.857$ ($p < 0.001$). The stripped "minimal" template induces task failure rather than rank reversal, consistent with insufficient-context degradation rather than construct invalidity. Rankings on

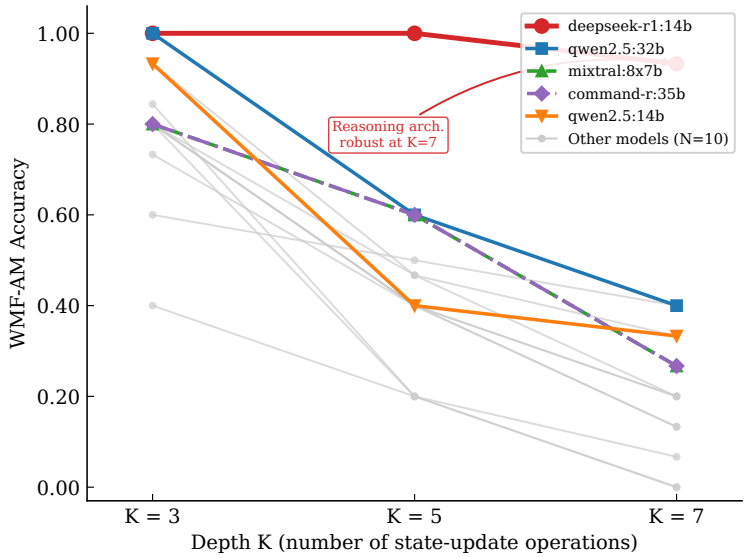

Figure 2. WMF-AM accuracy by depth K (N=15, chat template). deepseek-r1:14b (red) maintains near-perfect accuracy; all others degrade sharply at $K \geq 5$. Top 5 models highlighted.

the paraphrase-original template correlate near-perfectly with the main WMF-AM results: $\tau = 0.951$ ($p = 0.004$, N=7 overlapping models). (iii) Completed in prior loops: three surface forms (points, inventory, accounts), three prompt wrappers (bare, chat, CoT), and four random seeds, all showing stable inter-model rank orderings.

4.3 Predictor Comparison and Exploratory Results

Table 4 shows WMF-AM against five predictors of ABS (see Appendix S for the full 10-task table and battery release). The yoked cancellation control (arithmetic parsing only: τ=−0.031, p=0.879) supports the interpretation that the predictive signal is specific to cumulative state-tracking rather than arithmetic parsing ($\Delta\tau$=0.491, p=0.022). Exploratory partial τ controlling for OC (0.411, p=0.011) and scale (0.503, p<0.01) are hypothesis-generating; full details in Appendix Q.

Table 4. Predictor comparison: WMF-AM Score vs. alternative predictors of ABS. C = pre-specified; E = exploratory. $\Delta\tau$ (WMF vs. Yoked): 0.491 (p=0.022, bootstrap, inferentially confirmed). $\Delta\tau$ (WMF vs. OC/params): directional but not confirmed (95% CI overlaps zero).

Predictor	τ	p	N	Note
WMF-AM (this paper)	0.612	<0.001	20	C
log(params)	0.444	0.010	20	E
Outcome Correctness (OC)	0.428	0.012	20	E
K=1 single-step accuracy	0.411	0.047	15	E
Yoked cancellation accuracy	−0.031	0.879	15	E (parsing only)

4.4 Measurement Robustness

Multi-seed reliability: all 15 models evaluated with 4 independent seeds × 15 probes. Expansion-model SDs range 0.073–0.144 (mean 0.108), comparable to original-model variability. Cross-template stability: $\tau_{\text{bare,chat}} = 0.524$ ($p = 0.006$); $\tau_{\text{chat,cot}} = 0.314$ ($p = 0.114$). Leave-one-family-out (exploratory): τ ranges 0.503–0.649 across family removals (mean 0.588); no single family drives the effect. Family-aggregated (exploratory): Per-family median correlation $\tau = 0.538$ ($p = 0.030$, 13 families), suggesting the relationship holds at the family level. Full analyses in Appendix M.

Complementary probes (preliminary; Appendix N–P). Three probes at pilot scale ($N \leq 8$, single seed): EMC-lite (τ(WMF-AM)=0.250, consistent with independence); MCC-CE-v2 (error rates 23–40%, but 7/8 models show zero self-flagging); CLA-DC (collapses onto WMF-AM, τ=0.905, N=7). No multi-dimensional structure claims are drawn from these pilots.

4.5 Validity Summary

Appendix Q gives the complete validity register. (Divergent, E) WMF-AM ↔ OC τ=0.279 (p=0.150, N=15); WMF−Yoked vs OC τ=0.125 (p=0.519), consistent with process–outcome independence. (Convergent, E) CONV-RFPOC vs AGENT-PQ τ=0.905 (N=7; LLM judge; requires human annotation). (Template stability, C) bare↔chat τ=0.524 (p=0.006). Cross-dimension: CLA-DC collapses onto WMF-AM (τ=0.905); MCC-MA and EMC-lite remain separable.

5 Limitations, Alternative Views, and Discussion

Comparison to Minerva and K-calibration. Minerva [Xia et al., 2025] includes a Quantity State task at K=200; all 20 study models score exactly 0.000, rendering τ undefined. WMF-AM's K=3/5/7 calibration targets the discriminative window (scores span $[0.050, 0.983]$; τ=0.619 at K=3, 0.509 at K=7, undefined at K=200; Appendix B). Beyond K-calibration, WMF-AM adds a construct-isolation ablation suite.

Key limitations. Limitations are organized by severity into four buckets.

(L1) External validity (primary limitation). The sample comprises 20 open-weight models (0.5B–35B, greedy decoding, Ollama variants); claims do not extend to closed API models, fine-tuned variants, or non-English tasks. A single held-out check (llama3.1:70b; τ=0.627, N=21) is consistent with but does not constitute replication. The agent battery uses a single seed; multi-seed ABS reliability is not established. Next step: extend to $N \geq 30$ including API models.

(L2) Construct validity. The K=1 control separates cumulative tracking from single-step arithmetic; CoT collapses all models to ceiling (≥ 0.93), a construct boundary. Rival explanations (instruction-following, tokenizer effects) are mitigated by cross-template stability (τ=0.524) but not eliminated (Appendix J). A parsing artifact affects two verbose models (deepseek-r1:14b/7b); 88/90 responses contain the correct result.

(L3) Sample size and dependence structure. N=20 supports the two pre-specified primary analyses but not factor analysis; the four-dimension CEF structure is a provisional taxonomy, not a validated empirical model. Family non-independence is substantial (6 Qwen, 3 DeepSeek models); within-family ordering may trivially inflate τ. The within-Qwen robustness check (τ=0.894, p=0.016, N=6) is included but family-balanced sampling is needed for full rigor.

(L4) Comparator scope. The completion proxy is a custom 100-item battery, not a standard benchmark (MMLU, GSM8K). Partial-τ results controlling for completion or scale are exploratory; held-out replication is needed.

What WMF-AM is not. Ablations indicate WMF-AM is unlikely to primarily measure general arithmetic skill or working-memory capacity (K=1 control), generic entity tracking (non-arithmetic ceiling), or arithmetic parsing (yoked $\tau \approx 0$). Instruction-following and tokenizer effects are mitigated by cross-template stability but not eliminated; the mechanism linking cumulative tracking difficulty to diverse agent tasks is not established.

Scope of contribution. WMF-AM is validated in one regime: open-weight models (0.5B–35B), greedy decoding, no scratchpad. The two pre-specified findings (τ=0.612; τ=0.524) are in-sample evidence for this slice of model space; they do not generalize to process-sensitive probes at large. The CEF taxonomy is an organizing agenda, not an empirical result.

Implications. Evaluation suites would benefit from process probes alongside completion: a load-varied probe [Zhang et al., 2024], a metacognitive component [Lightman et al., 2023], and a difficulty gradient. Next steps: extend to API models ($N \geq 30$), test against standard benchmarks, and replicate across probe families.

Reproducibility. The 10-task agent battery, all WMF-AM probe templates, scoring scripts, per-model raw outputs, and the pre-registered analysis plan will be released upon publication at https://github.com/Hou-Dengzhe/CEF-WMF-AM.

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A Full Motivation and Related Work

Three failure modes of completion-only evaluation. (F1) Diagnostic blindness: models with near-identical completion scores can have very different process profiles. (F2) Uninformative failure diagnosis: current benchmarks report that an agent failed, not why, whether the bottleneck is state tracking, metacognitive calibration, or episodic confusion. (F3) Safety blindness: pattern-matching agents (which fail predictably at distribution shift) are indistinguishable from genuine reasoners on standard benchmarks.

Empirical support. Multiple realizability is empirically documented at scale. Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] show correct answers produced despite corrupted reasoning. Turpin et al. [Turpin et al., 2023] and Lanham et al. [Lanham et al., 2023] show CoT can be systematically unfaithful. PAE [Xie et al., 2026]: 27–78% of successful completions involve corrupted intermediate actions. AgentProcessBench [Fan et al., 2026]: 8,509-step human annotation shows weaker models inflate correct-step ratios via early termination. Chaduvula et al. [Chaduvula et al., 2026]: state-tracking inconsistency is 2.7× more prevalent in failed agent runs (TAU-bench).

Positioning relative to prior work. Entity Tracking [Kim and Schuster, 2023] is prior art for WMF-AM's design; Latent State Persistence [Huang et al., 2025] probes sequential latent-state manipulation on hidden integers, a closely related paradigm. Minerva [Xia et al., 2025]: see Table 5. CEF adapts cognitive science measurement logic [Miller, 1956,

Baddeley and Hitch, 1974, Flavell, 1979, Nelson and Narens, 1990, Tulving et al., 1972, Sweller, 1988] for API-level LLM evaluation without claiming LLMs have human cognition. Process supervision [Lightman et al., 2023, Uesato et al., 2022] trains models using step-level labels; CEF measures process quality. Mechanistic interpretability explains why a score is low; CEF identifies that it is low.

Table 5. Comparison to Minerva Quantity State. WMF-AM does not claim benchmark superiority; it contributes three additions to the existing arithmetic state-tracking paradigm.

Property	Minerva QS	WMF-AM (this paper)
Arithmetic state-tracking task	✓	✓
Step-count variation	✓(K variable)	✓(K=3, 5, 7)
Calibrated for open-weight discriminability	× (K=200: 20/20 models floor at 0.000)	✓(K=3/5/7: range [0.050, 0.983])
Construct-isolation ablation suite	×	✓(yoked, K=1, non-arith)
Downstream predictive validity	×	✓(τ=0.612, N=20)
Kendall τ with agent performance	undefined (constant)	0.612 (p<0.001)
Multi-seed + multi-template stability	×	✓
Head-to-head on same 20 models	N/A	✓(this paper)

B K-Sweep: Discriminability vs. Step Count

Table 6. K-sweep: discriminability vs. step count. τ=τ(probe, Agent); N = models with per-K data. K=3–7: WMF-AM per-depth means (N=12 models for which per-K breakdowns were collected; the remaining 8 were tested only at the composite level); K=20: 5-model inline test; K=200: full 20-model head-to-head.

K	N	Mean acc	Range	τ	p
3	12	0.512	$[0.050, 0.950]$	0.619	0.007
5	12	0.296	$[0.000, 1.000]$	0.545	0.020
7	12	0.192	$[0.000, 1.000]$	0.509	0.032
20	5	0.180	$[0.000, 0.900]$	-0.333	0.47
200 (Minerva)	20	0.000	$[0.000, 0.000]$	undef	undef

C CEF Framework Background

CEF (Cognitive Evaluation Framework) organizes process-sensitive probes into four proposed dimensions: WMF (Working Memory Fidelity, 3 sub-dimensions), MCC (Metacognitive Calibration, 2), EMC (Episodic Memory Coherence, 3), and CLA (Cognitive Load Adaptation, 3). Each dimension has a cognitive science anchor, a documented evaluation gap, and API-level testability. Labels like "working memory" are operational shorthand, not mechanistic claims (see Appendix G). Figure 3 illustrates the taxonomy and Table 7 summarizes the four dimensions.

Other CEF dimensions (background; not validated in this paper). MCC: decomposes metacognition into monitoring accuracy (MCC-MA) and control efficacy (MCC-CE). EMC: tests temporal ordering, cross-episode discrimination, and source attribution under interference. CLA: characterizes the performance-versus-load function shape; CLA-DC provisionally collapses onto WMF-AM (τ=0.905, N=7) pending factor analysis at $N\geq$20.

Construct validity framework. Following Cronbach and Meehl [1955], Messick [1989], Campbell and Fiske [1959], and Jacobs and Wallach [2021], each dimension requires convergent validity (correlation with independent process-sensitive measures) and divergent validity (low correlation with completion). The study supports divergent validity for WMF-AM (τ = 0.279 with a 100-item completion battery, N=15). CEF would be falsified if all dimensions collapsed onto a single factor indistinguishable from task completion. Figure 4 visualizes the validity evidence across probes, and Appendix I provides full construct-to-probe mapping.

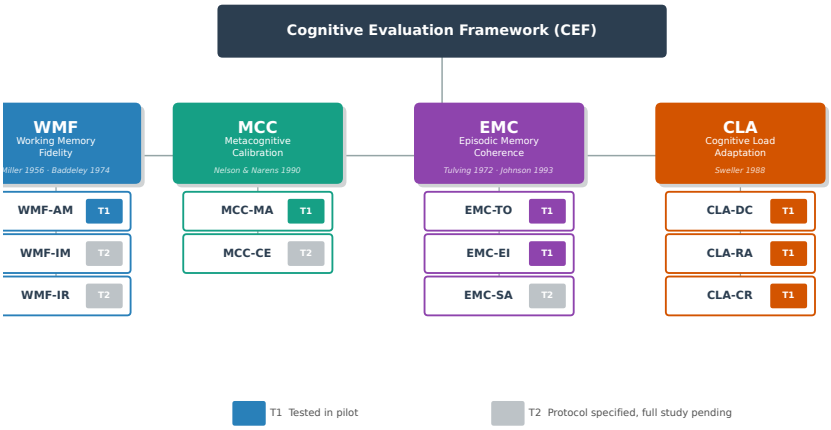

Figure 3. CEF taxonomy. Four proposed dimensions (WMF, MCC, EMC, CLA) with sub-dimensions, cognitive anchors, and probe mappings. Only WMF-AM is empirically validated in this paper; other dimensions are included as design context.

D Full Related Work Table

E Development Status and Summary Tables

F Glossary

G On Cognitive-Science Labels

We use labels like "working memory" and "metacognition" as operational shorthand for specific behavioral probes, not as claims about LLM internal mechanisms. "WMF-AM" means: "accuracy on sequential state-tracking tasks degrades with depth K"; it does not mean LLMs have a working memory buffer. "MCC-MA" means: "accuracy of self-error prediction on a 20-item probe"; it does not mean LLMs have phenomenal metacognition. The labels are justified because the probes are adapted from validated human measurement paradigms, and they organize what would otherwise be an unstructured collection of tests.

Two caveats are essential. First, we do not require LLMs to have phenomenal cognition or internal states analogous to human working memory. CEF dimensions are process-sensitive behavioral probes that borrow measurement logic from cognitive science. Second, CEF measurements are behavioral (prompt/response), designed to be differentially sensitive to process quality: high MCC-MA requires accurate self-error prediction; graceful CLA-DC requires proportional rather than catastrophic degradation.

H Full Sub-dimension Definitions

WMF-AM (Active Manipulation, Tier 1). Uses a complex span task: initial state, K sequential modifications, report final state. Depths $K \in \{3, 5, 7\}$ (dissociation study) and $K \in \{2, 4, 6, 8, 12\}$ (yoked cancellation control), three surface forms. Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential transformation.

WMF-IM (Information Maintenance, Tier 2). Embeds N facts in a passage with M distractors; measures recall accuracy. Extends Wang and Sun [Wang and Sun, 2025] to natural-language multi-turn contexts.

WMF-IR (Interference Resistance, Tier 2). Adapts the List-A/List-B proactive interference paradigm. Currently shows ceiling (= 1.0 all models); needs redesign.

MCC-MA (Monitoring Accuracy, Tier 1). Model predicts which of 20 answers are incorrect; Pearson r measures monitoring accuracy. Unlike Gnosis [Ghasemabadi and Niu, 2025] (requires internals), MCC-MA uses prompting under multi-turn load.

Table 7. CEF dimensions: target, failure signature, and exclusions. Detailed sub-dimension definitions appear in Appendix H.

Dim	Target	Failure signature	Excluded
WMF	Active maintenance and transformation of state under load	Accuracy drops as concurrent items exceed threshold	Long-context retrieval; knowledge recall
MCC	Self-monitoring of errors (MA) and correction of flagged errors (CE)	Confident-wrong; flagging without correction	General calibration (ECE); cross-model detection
EMC	Temporal ordering, cross-episode discrimination, source attribution	Episode blending; source misattribution	Single-turn retrieval; semantic memory
CLA	Degradation shape and resource scaling under load	Cliff-edge failure; non-recovery	Knowledge difficulty; unrelated complexity

Table 8. Representative benchmarks and evaluation work, grouped by measurement type and CEF gap addressed.

Work	Measures	CEF gap addressed
MMLU [Hendrycks et al., 2021a], ARC [Clark et al., 2018], HellaSwag [Zellers et al., 2019]	Knowledge recall	No memory structure, no process
HumanEval [Chen et al., 2021], MATH [Hendrycks et al., 2021b], GSM8K [Cobbe et al., 2021]	Output correctness	No reasoning fidelity
HotpotQA [Yang et al., 2018], WebShop [Yao et al., 2022]	Task completion	No metacognition, no memory type differentiation
AgentBench [Liu et al., 2023]	Sequential task success	No within-task process
Reflexion [Shinn et al., 2023]	Post-reflection improvement	Monitoring $\neq$ control (conflated)
PAE [Xie et al., 2026]	Corruption rate	Motivates CEF; no per-dimension measurement
NeuroCognition [Haznitrama et al., 2026]	Cognitive ability (IQ)	Ability $\neq$ process quality; no load variation
Strong Mem. & Weak Ctrl. [de Langis et al., 2025]	Executive function	Isolated tasks; no multi-turn load
Entity Tracking [Kim and Schuster, 2023]	Entity-state	Prior art for WMF-AM design (passive retrieval)
Latent State Persistence [Huang et al., 2025]	Sequential latent-state manipulation	Closest prior paradigm; no arithmetic load, no agent battery
Minerva [Xia et al., 2025]	Programmable memory (incl. Quantity State: sequential +/-)	Closest arithmetic prior; WMF-AM adds construct-isolation suite + agent-predictive validity
Zhang et al. [Zhang et al., 2024]	WM-style probes $\rightarrow$ reasoning limits	Motivational; no ablation suite, no agent battery
Intell. Degradation [Wang et al., 2026]	Load degradation	Prior art for CLA
HELM [Liang et al., 2023]	Behavioral outcomes	No cognitive processes
Long-context [Bai et al., 2024]	Retrieval from static window	WMF-AM targets active transformation
ECE calibration [Guo et al., 2017]	Confidence scalar	MCC decomposes into monitoring/control

MCC-CE (Control Efficacy, Tier 1). Model flags uncertain answers and revises them. Decomposes into flagging rate, correction efficacy, and false alarm rate.

EMC-TO/EI/SA. Temporal ordering (Kendall τ), cross-episode discrimination under similarity, and joint source-episode attribution. All proposed, no pilot data.

CLA-DC (Degradation Curve, Tier 1). Smoothness score (SD of successive Δs). Provisionally collapsed onto WMF-AM ($\tau = 0.905$).

CLA-RA/CR. Resource allocation (difficulty–elaboration correlation) and capacity recovery (post-load accuracy recovery). CLA-RA has no pilot data; CLA-CR-v3 spread = 1.171.

Composites (all exploratory). WMF = $0.50\times$AM$+0.30\times$IM$+0.20\times$IR; MCC = $0.55\times$MA$+0.45\times$CE; EMC = $0.40\times$EI$+0.35\times$TO$+0.25\times$SA; CLA = $0.50\times$DC$+0.30\times$RA$+0.20\times$CR. We do not propose an overall CEF composite; per-dimension profiles are the primary unit of analysis.

I Construct-to-Probe Mapping and Falsifiability

The framework as a whole would be undermined if process-sensitive measures never provided incremental prediction beyond completion on any downstream criterion. Table 12 specifies per-probe falsification conditions.

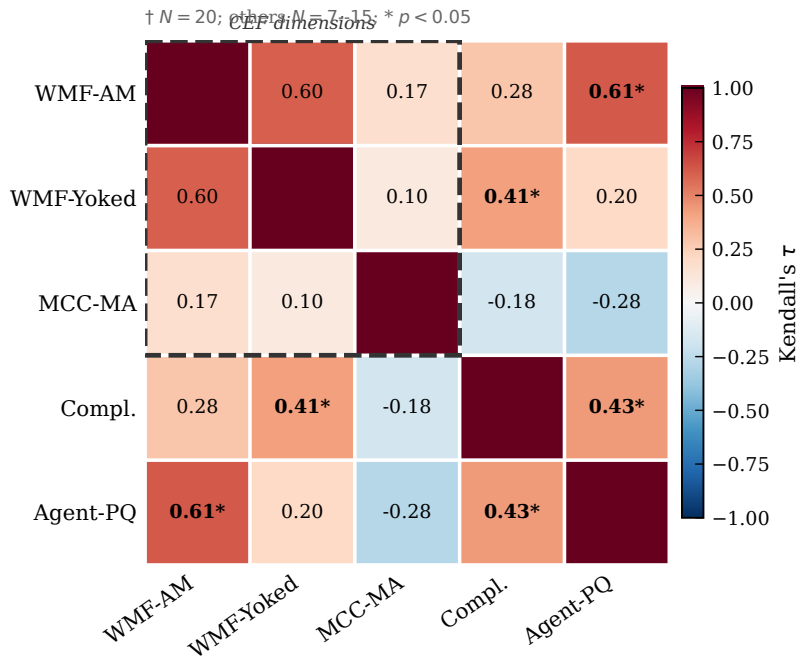

Figure 4. Validity evidence heatmap. Convergent and divergent correlations across CEF probes and external benchmarks (N=15). WMF-AM shows high divergent validity (low correlation with completion) and moderate convergent validity with agent battery scores.

Table 9. Development status of all 11 CEF sub-dimensions. Status key A: robust differentiation + multi-seed reliability; B: differentiation shown, limited seeds/models; C: probe redesigned, initial results; D: ceiling/floor effect, needs redesign; E: protocol specified, no pilot data.

Sub-dim	Status	Evidence	Limitation
WMF-AM	A^+	Δ=0.933, 4-seed all 20, 3-template (τ=0.524), yoked control, predictive τ=0.612	All open-weight
WMF-IM	B	spread 0.480, N=5	Single paradigm (hard)
WMF-IR	D	Ceiling (= 1.0 all models)	Needs redesign
MCC-MA	B	4/7 models differentiated	Floor for 3 models
MCC-CE	C	v2: error 23–40%, near-zero flagging (N=8)	Binary protocol
EMC-lite	B	Source-conflict spread 0.400	Single seed
CLA-DC	$B^\dagger$	Correlates τ=0.905 with WMF-AM	†Provisionally collapsed onto WMF
CLA-CR	C	v3 spread 1.171, N=7	v1–v2 failed; 2 seeds
CLA-RA	E	—	Awaits full study
EMC-TO	E	—	Awaits full study
EMC-EI	E	—	Awaits full study

J Rival Explanations

For each probe, plausible alternative explanations remain open. WMF-AM: (a) Instruction following: models may fail at depth because they lose track of instructions. The depth gradient partly addresses this. (b) Long-context retrieval: the control addresses this for 5/7 models. (c) Template artifacts: cross-template stability (bare↔chat τ=0.524, p=0.006; paraphrase original-vs-formal τ=0.857) mitigates but does not eliminate this. MCC-MA/CE: (a) Calibration shortcuts. (b) Prompt sensitivity. EMC-lite: Recency bias; keyword matching. These are the primary targets for the $N \geq 15$ study.

K Study Matrix

L Yoked Cancellation Control Details

The yoked cancellation control uses operation pairs that cancel: "Alice gains 3 points" followed immediately by "Alice loses 3 points." The prompt format, entity names, and depth structure match WMF-AM exactly, so the arithmetic parsing demand is identical. The correct answer equals the initial state, but unlike the original inert control, the model

Table 10. CEF at a glance. Each sub-dimension's task type, scoring rule, and pilot status. Tier 1 = novel construct; Tier 2 = novel reframing. Pilot: ✓ = pilot-supported, ∼ = preliminary, — = protocol only.

Sub-dim	Task (one-line)	Scoring	Tier	Pilot
WMF-AM	Track K sequential state updates, report final state	Exact match	1	✓
WMF-IM	Recall N facts amid M distractors	Recall accuracy	2	∼
WMF-IR	Recall List-A after List-B interference	Proactive interference δ	2	—
MCC-MA	Predict which of 20 answers are wrong	Pearson r (predicted vs. actual)	1	∼
MCC-CE	Flag uncertain answers, revise them	Correction efficacy	1	∼
EMC-TO	Reorder non-chronological events	Kendall τ vs. ground truth	1	—
EMC-EI	Discriminate similar episodes	Accuracy under similarity	1	—
EMC-SA	Attribute facts to source + episode	Joint attribution accuracy	2	—
CLA-DC	Performance across 5 load levels	Smoothness (SD of Δs)	1	—
CLA-RA	Response elaboration vs. difficulty	Pearson r	1	—
CLA-CR	Easy-task recovery after high-load block	Recovery rate	1	∼

Table 11. Glossary of key terms and acronyms.

Term	Definition
CEF	Cognitive Evaluation Framework (this paper)
WMF-AM/IM/IR	Working Memory Fidelity: Active Manipulation / Information Maintenance / Interference Resistance
MCC-MA/CE	Metacognitive Calibration: Monitoring Accuracy / Control Efficacy
EMC-TO/EI/SA	Episodic Memory Coherence: Temporal Ordering / Episodic Interference / Source Attribution
CLA-DC/RA/CR	Cognitive Load Adaptation: Degradation Curve / Resource Allocation / Capacity Recovery
AGENT-PQ/TC	Agent Process Quality / Task Completion (external evaluation)
CONV-RFPOC	Convergent validity probe: Reasoning Fidelity–Product of Cognition

must process each operation to verify cancellation. This eliminates the trivial shortcut of ignoring operations.

M Measurement Robustness

Multi-seed reliability. All 15 models evaluated with 4 independent random seeds × 15 probes. Original-model cross-seed $\tau = 0.798$. Expansion-model SDs: phi3:14b = 0.094, gemma2:9b = 0.144, qwen2.5:3b = 0.122, llama3.2:3b = 0.122, deepseek-r1:7b = 0.114, mixtral:8x7b = 0.116, command-r:35b = 0.100, yi:34b = 0.084 (mean SD = 0.108).

Cross-template stability. Phase 5 administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) across all 15 models. Bare ↔ chat: $\tau = 0.524$ ($p = 0.006$), significant rank preservation. Bare ↔ CoT: $\tau = 0.105$ ($p = 0.627$), CoT pushes all models to ceiling (≥ 0.93), eliminating variance. Chat ↔ CoT: $\tau = 0.314$ ($p = 0.114$). The CoT ceiling effect is interpretively meaningful: externalized reasoning via scratchpad bypasses the working-memory bottleneck that WMF-AM is designed to measure.

Non-ceiling completion proxies. τ(WMF-AM, MMLU) = 0.356 ($p = 0.317$; range 0.25–0.72); τ(WMF-AM, GSM8K) = 0.150 ($p = 0.645$; range 0.05–0.83). Scores from published model cards, not re-evaluated under pilot setup.

Depth profile. See Figure 2 in the main text for depth-dependent degradation curves.

N EMC-lite Results

EMC-lite provides preliminary support for the source-conflict slice. It does not establish EMC as a whole; single-seed evidence only. τ(WMF-AM, EMC-lite) = 0.250: substantial independence.

O MCC-CE-v2 Results

The v1 floor was an artifact of easy questions; v2 models make many errors but show near-zero self-flagging under the binary protocol. Alternative designs (continuous confidence,

Table 12. Construct map for validated probes: manipulated factor, observable signature, rival explanations, and falsification criteria.

Probe	Manipulated factor	Observable signature	Plausible rivals	Expected pattern	Falsified if
WMF-AM	Depth K (number of sequential state updates)	Accuracy × depth interaction	Instruction following; tokenizer parsing; template familiarity; long-context retrieval	Low τ with completion; monotonic depth decline; $\tau_{\text{bare,chat}} > 0.4$; yoked control $\gg$ WMF-AM	High τ with completion ($\geq$ 0.55); no depth effect; template-specific rankings; control degrades at same rate
MCC-MA	Error-laden vs correct items	Pearson r (predicted vs actual errors)	Calibration shortcuts; always-predict-correct bias	Low τ with WMF-AM; model-specific prediction	Uniform predictions; $\tau \approx 1$ with WMF-AM
EMC-lite	Conflict vs control episodes	Source attribution accuracy under interference	Recency bias; keyword matching	Low τ with WMF-AM; fault > control difficulty	No fault–control difference; $\tau \approx 1$ with WMF-AM

Table 13. Study matrix: all experiments, models, and sample sizes. †llama3.1:70b added for MCC-CE-v2 only.

Experiment	Metric	N_{models}	Seeds	Trials/model
Outcome Correctness (v2)	Accuracy	20	1	100
WMF-AM (dissociation)	Exact match	20	4	180
Agent Validation	Deterministic task score	20	1	10
Template Harmonization	Rank stability	15	1	135
Yoked Cancellation Control	Exact match	15	1	100
MCC-MA	Pearson r	15	1	20
MCC-CE-v2	Flagging rate	8†	1	50
EMC-lite	Composite	7	1	18
CLA-CR-v3	Recovery rate	7	2	7 blocks
AGENT-PQ/TC	Rubric / binary	7	1	10
CONV-RFPOC	Fidelity	7	1	15

forced ranking) might reveal partial capacity.

P Negative OOD Result

Two OOD stress-test designs (v1: 5-step NL tasks; v2: K=8–9 interference tasks) produced anti-correlated CEF–OOD relationships across model pairs. We make no predictive claim. Full assessment requires $N \geq 11$ with process-sensitive downstream tasks.

Q Full Validity Analyses

Below we provide additional details on the validity analyses summarized in Section 4.5.

Convergent validity details. AGENT-PQ: 10 multi-step scenarios scored by LLM judge (GPT-4o) on 4 rubric dimensions. CONV-RFPOC: 15 reasoning problems with counterfactual chain perturbation. RFPOC $= 1 - P(\text{same answer} \mid \text{perturbed chain})$. Full protocol in Appendix R. Limitation: LLM judge may introduce shared bias.

Construct separation (WMF-AM $\leftrightarrow$ CLA-DC). $\tau = 0.905$: no evidence of separability at N=7. Both involve state tracking under load. Planned discriminant test at $N \geq 15$. CLA-CR-v3 provides a structurally independent CLA measure. Even if CLA-DC merges with WMF-AM, WMF-AM, MCC-MA, and EMC-lite remain separable ($\tau \leq 0.250$).

Sensitivity. LOMO for CONV-RFPOC vs AGENT-PQ: τ ranges 0.600–1.000 (mean 0.833). Leave-one-family-out (Qwen, N=4): $\tau = 0.667$. After Holm correction, RFPOC–AGENT-PQ remains significant ($p_{\text{adj}} = 0.021$).

Multiple testing. Two primary hypotheses; all others exploratory.

Table 14. EMC-lite (N=7, hardened protocol). Source attribution shows strongest differentiation (spread = 0.500).

Model	F-Det	F-Src	F-Rec	F-Comp	C-Rec	ΔRec
qwen2.5:14b	0.917	0.750	0.917	0.917	0.972	−0.056
qwen2.5:32b	0.917	0.750	0.917	0.917	1.000	−0.083
deepseek-r1:14b	0.889	0.917	0.917	0.912	1.000	−0.083
llama3.1:8b	0.778	0.583	0.750	0.771	0.889	−0.139
qwen2.5:7b	0.861	0.583	0.639	0.736	0.722	−0.083
mistral:7b	0.750	0.417	0.639	0.685	0.722	−0.083
gemma2:27b	0.861	0.583	0.528	0.667	0.611	−0.083
Δ	0.167	0.500	0.389	0.250	0.389	—

Table 15. MCC-CE-v2 (N=8). Error rates 23–40% (floor eliminated); flagging near zero for all models.

	Balanced (n=30)		Trap-only (n=20)	
Model	Error%	Flag%	Error%	Flag%
qwen2.5:7b	26.7	0.0	40.0	0.0
qwen2.5:14b	36.7	0.0	40.0	0.0
qwen2.5:32b	30.0	0.0	25.0	0.0
llama3.1:8b	26.7	0.0	40.0	0.0
llama3.1:70b	—	—	25.0	0.0
gemma2:27b	30.0	11.1	30.0	0.0
deepseek-r1:14b	23.3	0.0	35.0	0.0
mistral:7b	30.0	0.0	40.0	0.0

Benchmark governance. All headline results use frozen protocol versions. Probe revisions were motivated by floor/ceiling effects, not model-specific tuning.

R AGENT-PQ Evaluation Protocol

AGENT-PQ uses 10 multi-step scenarios where each model operates within a tool-use scaffold. A separate LLM judge (GPT-4o, temperature = 0) scores each trajectory on four rubric dimensions: plan coherence, state tracking, error recovery, goal adherence. AGENT-PQ = mean of 4 rubric scores/4. AGENT-TC is the binary success rate on the same scenarios.

CONV-RFPOC administers 15 reasoning problems with counterfactual chain perturbation. RFPOC $= 1 - P(\text{same answer} \mid \text{perturbed chain})$.

S Complete Experimental Protocols

Full prompt templates, parameter specifications, scoring procedures, and reference implementations for all 11 sub-dimensions are provided in the supplementary materials.

T Weight Sensitivity Analysis

Three weighting schemes (proposed, equal, PCA-derived) are evaluated. Per-dimension profiles are primary; composites are exploratory only.

U Dissociation Case Studies

Matched dissociation pairs: deepseek-r1:14b and qwen2.5:14b share similar EMC-lite (0.912 vs. 0.917) but differ on WMF-AM (0.983 vs. 0.467). mistral:7b and llama3.1:8b have similar WMF-AM but differ on EMC-lite. These cross-dimension pairs anchor the multiple realizability argument.

V Worked Example: WMF-AM at Depth K=3

Setup. "A warehouse has 3 items: Widget=50, Gadget=30, Sprocket=20." Three updates: "Ship 10 Widgets," "Receive 25 Gadgets," "Transfer 5 Sprockets to overflow."
Correct. Widget=40, Gadget=55, Sprocket=15.
Scoring. Exact match per item. deepseek-r1:14b scores 1.000 at K=7; llama3.1:8b scores 0.050. Both score 0.90–0.95 on outcome correctness. This illustrates the Completion Fallacy.

W Novelty Classification

Table 16. Novelty tier for all 11 sub-dimensions. Tier 1 = novel; Tier 2 = novel reframing.

Sub-dim	Tier	Prior art / differentiation
WMF-AM	1	Entity Tracking [Kim and Schuster, 2023]: passive retrieval; Latent State Persistence [Huang et al., 2025]: structural ops, no arithmetic load, no agent battery; WMF-AM = cumulative arithmetic transformation + ablation suite + downstream validity
WMF-IM	2	Wang & Sun [Wang and Sun, 2025]: key-value pairs; WMF-IM = NL multi-turn
WMF-IR	2	Extended to domain-varied materials
MCC-MA	1	Gnosis [Ghasemabadi and Niu, 2025] requires internals; MCC-MA = prompting only
MCC-CE	1	Reflexion [Shinn et al., 2023]: aggregate; MCC-CE decomposes detection/correction
EMC-TO	1	No prior LLM temporal ordering measurement
EMC-EI	1	No prior cross-episode interference diagnostic
EMC-SA	2	Interference-conditioned attribution
CLA-DC	1	Intell. Degradation [Wang et al., 2026]: magnitude; CLA-DC = shape
CLA-RA	1	Not previously measured
CLA-CR	1	Not previously measured